

RESTful API Documentation & Interface Specification

Comprehensive OpenAPI Specification for Catalog, Bookings, ML Inference & AI Endpoints

Production documentation artifact for the AI-Powered Household Appliance Rental System (RentAI / SDC2), encompassing enterprise Pure NoSQL architecture, customer churn prevention, multi-paradigm collaborative filtering, and local LLM context assistance.

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Project Title:	RentAI / SDC2 Household Appliance Ecosystem	Classification:	Confidential / Academic & Enterprise
Lead Architecture:	Deep Learning & Full-Stack Systems Team	Publication Date:	September 2026
Status:	VERIFIED & APPROVED	Target Platform:	Ubuntu 24.04 LTS / Windows 11 Enterprise

1. API Architecture & Standards

RentAI's API adheres to RESTful conventions over HTTP/1.1 and HTTP/2. All payloads are encoded in standard UTF-8 JSON. Authentication uses JWT Bearer tokens passed via the `Authorization` header.

2. Endpoint Directory & Contract Details

2.1 GET `/api/appliances/` — Appliance Catalog

Retrieves a filtered, paginated list of available appliances.

Query Parameters: `category` (str), `max_price` (float), `available` (bool), `search` (str).

Response (200 OK):

```
[
  {
    "id": "66f4b81c2f1a9b0012345678",
    "name": "LG 260L Frost-Free Double Door Refrigerator",
    "category": "Refrigerator",
    "brand": "LG",
    "price_per_day": 210.0,
    "monthly_rent": 6300.0,
    "deposit": 12000.0,
    "available": true,
    "location": "Hyderabad"
  }
]
```

2.2 POST `/api/chat/` — Local Contextual RAG Chat

Dispatches natural language query to local Ollama inference server with MongoDB RAG context.

Request Body: `{"message": "Show fridges under 300", "stream": false, "model": "rentai-llm"}`

Response (200 OK):

```
{
  "response": "We have the LG 260L Refrigerator at ₹210/day and Whirlpool 190L at ₹180/day in H",
  "model": "rentai-llm:latest",
  "context_used": 3
}
```

2.3 GET `/api/chat/status/` — Ollama Health & Model Discovery

Polls port 11434, auto-starts the Ollama background daemon if offline, and returns detected local models.

Response (200 OK):

```
{
  "connected": true,
  "status": "ready",
  "active_model": "rentai-llm:latest",
  "models": [
    {"name": "qwen2.5:0.5b", "size_mb": 379.4, "param_size": "494.03M"},
    {"name": "rentai-llm:latest", "size_mb": 1259.9, "param_size": "1.2B"}
  ],
  "device": "CPU (Optimized)"
}
```

2.4 GET `/api/churn/at-risk/` — LightGBM High-Risk Churn Detection

Scans active tenant population, executing the 15-feature LightGBM classification pipeline.

Response (200 OK): Returns array of tenants with `churn_probability ≥ 0.65` and risk indicators.